Exercise Set 11: Lie Groups and Lie Algebras

A Lie group is a smooth manifold that is also a group, with the group operations smooth maps. Its tangent space at the identity is the ***Lie algebra*** $\mathfrak{g}$*, which encodes the infinitesimal structure of the group. Every physical symmetry is a Lie group; every conservation law (via Noether's theorem from Set 7) corresponds to an element of the Lie algebra. The deepest example is spin: the rotation group* $SO(3)$ *and its double cover* $SU(2)$ *have the same Lie algebra but different topology, and this difference is why electrons exist.*

Recall of Definitions and Examples

Lie groups

R1. Write the definition of a **Lie group** G. What two structures must it carry simultaneously? What smoothness conditions are required of the group operations?

R2. For each of the following, verify it is a Lie group by identifying the group operation and the smooth manifold structure:

(a) $(\mathbb{R}^n, +)$

(b) $(S^1, \cdot)$ the circle group

(c) $GL(n, \mathbb{R})$ invertible $n \times n$ real matrices

(d) $SO(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I, \det A = 1\}$

(e) $SU(n) = \{A \in GL(n, \mathbb{C}) : A^\dagger A = I, \det A = 1\}$

(f) $U(n) = \{A \in GL(n, \mathbb{C}) : A^\dagger A = I\}$

R3. (Dimensions Nawata HW9v2 Problem 1.) Compute the real dimension of each Lie group by counting the degrees of freedom in the defining matrix equation.

(a) $GL(n, \mathbb{R})$: $\dim = ?$

(b) $SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A = 1\}$: how many constraints does $\det A = 1$ impose?

(c) $O(n)$: the condition $A^T A = I$ is a symmetric matrix equation. How many independent constraints? What is $\dim O(n)$?

(d) $SO(n)$: same as $O(n)$ but connected. $\dim SO(n) = ?$

(e) $U(n)$: $A^\dagger A = I$ count real constraints. $\dim U(n) = ?$

(f) $SU(n)$: $\dim SU(n) = ?$

(g) $Sp(n) = \{A \in GL(2n, \mathbb{R}) : A^T J A = J\}$ where $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$: $\dim Sp(n) = ?$

Lie algebras

R4. Write the definition of the **Lie algebra** $\mathfrak{g}$ of a Lie group G in two equivalent ways:

(a) As the tangent space at the identity: $\mathfrak{g} = T_e G$

(b) As the space of left-invariant vector fields on G

Why are these equivalent? What is the Lie bracket on $\mathfrak{g}$?

R5. (Lie algebras explicitly Nawata HW9v2 Problem 2.) The Lie algebra of a matrix Lie group $G \subset GL(n)$ is:

$$\mathfrak{g} = \{X \in \mathfrak{gl}(n) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\}.$$

Use this to write the Lie algebras of each group:

(a) $\mathfrak{gl}(n, \mathbb{R})$: all $n \times n$ real matrices

(b) $\mathfrak{sl}(n, \mathbb{R}) = \{X : \mathrm{tr}(X) = 0\}$. Verify: if $\mathrm{tr}(X) = 0$ then $\det(e^{tX}) = e^{t\,\mathrm{tr}X} = 1$.

(c) $\mathfrak{so}(n) = \{X : X^T = -X\}$ antisymmetric matrices. Verify by differentiating $A^T A = I$ at $A = e^{tX}$.

(d) $\mathfrak{u}(n) = \{X : X^\dagger = -X\}$ skew-Hermitian matrices.

(e) $\mathfrak{su}(n) = \{X : X^\dagger = -X,\ \mathrm{tr}(X) = 0\}$.

For each, compute the real dimension and verify it matches the group dimension from R3.

R6. Write the definition of the **exponential map** $\exp : \mathfrak{g} \to G$. For matrix groups, what is the formula? What is $\exp(0)$? Is exp always surjective?

R7. State the **Baker-Campbell-Hausdorff formula** (first two terms):

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$$

When does $e^X e^Y = e^{X+Y}$ hold exactly?

Key examples

R8. Write a basis for $\mathfrak{so}(3)$ using the matrices J_x, J_y, J_z from Exercise Set 7 (C7). Write the commutation relations:

$$[J_x, J_y] = ?, \qquad [J_y, J_z] = ?, \qquad [J_z, J_x] = ?$$

R9. Write a basis for $\mathfrak{su}(2)$ using the Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Show the basis elements $e_i = \frac{i}{2}\sigma_i$ satisfy:

$$[e_x, e_y] = e_z, \qquad [e_y, e_z] = e_x, \qquad [e_z, e_x] = e_y.$$

Compare with $\mathfrak{so}(3)$. What do you notice?

Computation

C1. ($e^A e^B \neq e^{A+B}$ Nawata HW9v2 Problem 3.)

(a) Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Compute e^A, e^B, e^{A+B}, and $e^A e^B$ explicitly. Verify $e^A e^B \neq e^{A+B}$.

(b) Compute $[A, B] = AB - BA$. Then verify the BCH formula to second order: $e^A e^B \approx e^{A+B+\frac{1}{2}[A,B]}$.

(c) Show that $e^A e^B = e^{A+B}$ for all $A, B \in \mathfrak{g}$ if and only if G is abelian.

(d) **Correction formula.** Show that the exact equality is:

$$e^X e^Y = e^{X+Y} \iff [X, Y] = 0.$$

Give an example of commuting matrices where the equality holds.

C2. (The Lie algebra $\mathfrak{so}(3)$ and angular momentum.)

(a) The matrices

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{so}(3)$. Verify $J_i^T = -J_i$ (antisymmetric) and $\mathrm{tr}(J_i) = 0$.

(b) Compute the Lie brackets $[J_x, J_y]$, $[J_y, J_z]$, $[J_z, J_x]$.

(c) Compute $\exp(tJ_z)$ explicitly by summing the series.
Hint: $(J_z)^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $(J_z)^3 = -J_z$, *etc.* What rotation does $\exp(tJ_z)$ represent?

(d) Show that $T_e SO(3)$ is spanned by J_x, J_y, J_z by showing these are the tangent vectors to the curves $\exp(tJ_i)$ at $t = 0$. This is Nawata HW2 Problem 3.

(e) $\dim SO(3) = 3$. Match this with the computation in R3.

C3. (The Lie algebra $\mathfrak{su}(2)$ and Pauli matrices.)

(a) Show that $e_i = \frac{i}{2}\sigma_i$ are skew-Hermitian and traceless they lie in $\mathfrak{su}(2)$.

(b) Compute $\exp(2\pi e_z) = \exp(i\pi\sigma_z)$ explicitly. What is the result? Compare with $\exp(2\pi J_z) \in SO(3)$.

(c) Compute $\exp(4\pi e_z)$. What does this say about the geometry of $SU(2)$?

(d) $\dim SU(2) = 3 = \dim SO(3)$. The Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ are isomorphic (same commutation relations). But $SU(2) \ncong SO(3)$ as Lie groups. What is different?

C4. ($SO(4)$ and quaternions Nawata HW9 Problem 3.)

Recall from Set 5 (P4) that $SU(2) = \{q \in \mathbb{H} : |q| = 1\}$ is the group of unit quaternions.

(a) The map $\phi : SU(2) \times SU(2) \to SO(4)$ defined by $(q_1, q_2) \cdot x = q_1 x q_2^{-1}$ for $x \in \mathbb{H} \cong \mathbb{R}^4$ is a group homomorphism. Show it preserves the norm: $|q_1 x q_2^{-1}| = |x|$. Conclude ϕ maps into $O(4)$.

(b) Show the kernel of ϕ is $\{(\pm 1, \pm 1)\} \cong \mathbb{Z}_2$ (the pairs $(1,1)$ and $(-1,-1)$ give the same rotation).

(c) Conclude $SO(4) \cong (SU(2) \times SU(2))/\mathbb{Z}_2$.

(d) Compute $\dim SO(4)$ directly from R3 and verify: $\dim(SU(2) \times SU(2)) = 3 + 3 = 6 = \dim SO(4)$. ✓

(e) What is $\pi_1(SO(4))$?
Hint: $\pi_1(SU(2) \times SU(2)) = 0$ since $SU(2) \cong S^3$. Use the long exact sequence of the fibration $\mathbb{Z}_2 \to SU(2) \times SU(2) \to SO(4)$.

C5. (The Lorentz group Nawata HW9v2 Problem 4.)

Define $\sigma^\mu = (I, \sigma_x, \sigma_y, \sigma_z)$ and

$$X = x^\mu \sigma_\mu = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix} \in \mathrm{Herm}(2)$$

(a 2×2 Hermitian matrix). Note $\det X = t^2 - x^2 - y^2 - z^2$.

(a) Compute $\det X$ in terms of (t, x, y, z) and verify it equals the Minkowski norm $\eta_{\mu\nu} x^\mu x^\nu$.

(b) For $A \in SL(2, \mathbb{C})$, the map $X \mapsto AXA^\dagger$ preserves Hermiticity and satisfies $\det(AXA^\dagger) = \det X$. Show this defines a map $\phi : SL(2, \mathbb{C}) \to SO^+(1,3)$.

(c) The kernel of ϕ is $\{\pm I\}$. Conclude $SO^+(1,3) \cong SL(2, \mathbb{C})/\{\pm I\}$.

(d) Compare with $SO(3) \cong SU(2)/\{\pm I\}$: going from $\mathbb{R}^3$ to $\mathbb{R}^{1,3}$ replaces $SU(2)$ with $SL(2, \mathbb{C})$. Complete the table:

Rotation group	Double cover	Spacetime
$SO(3)$	$SU(2)$	$\mathbb{R}^3$ (Euclidean)
$SO^+(1,3)$	$SL(2, \mathbb{C})$	$\mathbb{R}^{1,3}$ (Minkowski)

(e) Compute $\pi_1(SO^+(1,3))$.
Hint: $SL(2,\mathbb{C})$ deformation retracts to $SU(2) \cong S^3$, which is simply connected.

Physics: Why Spin-$\frac{1}{2}$ Exists

The rotation group $SO(3)$ and the group $SU(2)$ have identical Lie algebras the same infinitesimal structure. But they are topologically different: $\pi_1(SO(3)) = \mathbb{Z}_2$ while $\pi_1(SU(2)) = 0$. This topological difference has a profound physical consequence: particles can transform under $SU(2)$ but not under $SO(3)$, giving rise to half-integer spin. Electrons, quarks, and neutrinos are all spinors representations of $SU(2)$, not $SO(3)$. Without the double cover structure of Lie groups, they could not exist.

P1. (The double cover $SU(2) \to SO(3)$.)

(a) From C3, we have $\exp(2\pi e_z) = -I \in SU(2)$ but $\exp(2\pi J_z) = I \in SO(3)$. This means a 360ř rotation maps to $-I$ in $SU(2)$ but back to I in $SO(3)$. Describe the covering map $\rho : SU(2) \to SO(3)$: what does $\rho(U)$ do to a vector $\vec{v} \in \mathbb{R}^3$?
Hint: identify $\mathbb{R}^3 \cong \mathfrak{su}(2)$ and let $\rho(U)(X) = UXU^\dagger$.

(b) Show the kernel of ρ is $\{\pm I\} \cong \mathbb{Z}_2$. Both U and $-U$ give the same rotation. This is the 2-to-1 covering.

(c) Show $SO(3) \cong SU(2)/\{\pm I\}$. Since $SU(2) \cong S^3$ (simply connected):

$$\pi_1(SO(3)) \cong \mathbb{Z}_2.$$

This means: in $SO(3)$, a 360ř rotation is not contractible to the identity, but a 720ř rotation is.

(d) **The Dirac belt trick revisited.** In Set 5 (P4) we saw the belt trick: 360ř twists the arm, 720ř untwists it. Now explain it precisely:

- A physical object (plate, electron) moving through a 360ř rotation traces a loop in $SO(3)$
- This loop is non-contractible ($\pi_1(SO(3)) = \mathbb{Z}_2$) the arm stays twisted
- A 720ř rotation traces a loop that goes around twice this IS contractible in $SO(3)$ the arm untwists

Draw the two loops in $SO(3) \cong \mathbb{R}P^3$ and indicate which is contractible.

P2. (Representations and spin.)

A **representation** of a Lie group G is a smooth homomorphism $\rho : G \to GL(V)$ for some vector space V. A **representation of the Lie algebra $\mathfrak{g}$** is a Lie algebra homomorphism $d\rho : \mathfrak{g} \to \mathfrak{gl}(V)$.

(a) The **standard representation** of $SO(3)$ is the 3×3 rotation matrices acting on $\mathbb{R}^3$. Under a 360ř rotation $R_{2\pi}$, what does every vector $\vec{v} \in \mathbb{R}^3$ map to? This is spin-1: $R_{2\pi} = +I$.

(b) The **fundamental representation** of $SU(2)$ is the 2×2 matrices acting on $\mathbb{C}^2$. Under a 360ř rotation $U_{2\pi} = \exp(2\pi e_z) = -I$, what does every vector $\psi \in \mathbb{C}^2$ map to? This is spin-$\frac{1}{2}$: $U_{2\pi} = -I$.

(c) A spin-$\frac{1}{2}$ particle is a physical state described by $\psi \in \mathbb{C}^2$ that transforms under a rotation $U \in SU(2)$ (not $R \in SO(3)$). Under 360ř: $\psi \mapsto -\psi$. Under 720ř: $\psi \mapsto +\psi$. This is physically observable the Aharonov-Bohm effect for neutrons confirms $\psi \mapsto -\psi$ under 360ř.

(d) The representations of $SU(2)$ are labelled by $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ The representation of spin j has dimension $2j + 1$ and $U_{2\pi} = (-1)^{2j} I$.

- $j = 0$: trivial representation, $U_{2\pi} = +I$ (scalar)
- $j = \frac{1}{2}$: fundamental, $U_{2\pi} = -I$ (electron)
- $j = 1$: adjoint, $U_{2\pi} = +I$ (photon)
- $j = \frac{3}{2}$: $U_{2\pi} = -I$ (delta baryon)

The half-integer spins ($j = \frac{1}{2}, \frac{3}{2}, \ldots$) are representations of $SU(2)$ that do *not* descend to representations of $SO(3)$. They can only exist because of the double cover structure.

(e) Write one sentence: why does the existence of electrons (spin-$\frac{1}{2}$) require $\pi_1(SO(3)) = \mathbb{Z}_2$ and not $\pi_1(SO(3)) = 0$?

P3. (Spinors and the Lorentz group.)

The same story repeats for the Lorentz group: $SO^+(1,3)$ has double cover $SL(2,\mathbb{C})$.

(a) $\pi_1(SO^+(1,3)) = \mathbb{Z}_2$ (from C5). What does this say about Lorentz boosts composed with rotations?

(b) A **Dirac spinor** $\psi \in \mathbb{C}^4$ transforms under the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation of $SL(2,\mathbb{C})$. Under a spatial 360ř rotation: $\psi \mapsto -\psi$. This is the Dirac equation the relativistic wave equation for spin-$\frac{1}{2}$ particles.

(c) The Dirac equation requires both $SL(2,\mathbb{C})$ (the double cover) and the specific representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. The $(\frac{1}{2}, 0)$ piece is a **left-handed Weyl spinor** and $(0, \frac{1}{2})$ is a **right-handed Weyl spinor**. Neutrinos in the Standard Model are left-handed Weyl spinors they only come in one chirality. Why does chirality require the Lorentz group and not just $SO(3)$?

(d) Complete the table:

Spacetime	Symmetry	Double cover	Spinors
$\mathbb{R}^3$ (non-relativistic)	$SO(3)$	$SU(2)$	$\mathbb{C}^2$ (Pauli spinor)
$\mathbb{R}^{1,3}$ (relativistic)	$SO^+(1,3)$	$SL(2,\mathbb{C})$	$\mathbb{C}^4$ (Dirac spinor)

P4. (The Lie algebra is not enough topology matters.)

Both $SO(3)$ and $SU(2)$ have the same Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$. The difference is global topology.

(a) As smooth manifolds: $SU(2) \cong S^3$ (simply connected) and $SO(3) \cong \mathbb{R}P^3$ (fundamental group $\mathbb{Z}_2$). Verify $\dim SU(2) = \dim SO(3) = 3$.

(b) The exponential map $\exp : \mathfrak{su}(2) \to SU(2)$ is surjective (every element of $SU(2)$ is $\exp(X)$ for some X). Is $\exp : \mathfrak{so}(3) \to SO(3)$ surjective?

(c) Two Lie groups can have the same Lie algebra but different global structure. The Lie algebra sees only the *local* neighbourhood of the identity. The *global* topology π_1, π_2, etc. is invisible to the Lie algebra.

(d) This is why the classification of Lie groups is harder than the classification of Lie algebras: given a Lie algebra $\mathfrak{g}$, there is a unique *simply connected* Lie group $\tilde{G}$ with $\mathrm{Lie}(\tilde{G}) = \mathfrak{g}$. All other groups with the same Lie algebra are quotients $\tilde{G}/\Gamma$ for discrete subgroups $\Gamma \subset Z(\tilde{G})$. For $\mathfrak{su}(2)$: $\tilde{G} = SU(2)$, $\Gamma = \mathbb{Z}_2$, giving $SO(3) = SU(2)/\mathbb{Z}_2$. Verify this matches everything computed above.

(e) **The physical moral.** Quantum mechanics allows particles to transform under *any* Lie group with the right Lie algebra not just the one physicists first write down. The correct group is determined by experiment: electrons transform under $SU(2)$, not $SO(3)$. Topology, not just algebra, determines the spectrum of particles.

The topological difference $\pi_1(SO(3)) = \mathbb{Z}_2$ vs $\pi_1(SU(2)) = 0$ has a direct physical consequence: half-integer spin particles (electrons, quarks) exist. They transform under $SU(2)$, picking up a sign under 360ř rotation. This sign is undetectable classically but measurable quantum mechanically.

The same story in spacetime: $SO^+(1,3)$ has double cover $SL(2,\mathbb{C})$, giving Dirac spinors the mathematical objects describing all matter particles in the Standard Model.

In the next sets we build fiber bundles and connections the geometric setting where gauge fields live and where the Dirac spinor finds its natural home.***Reflection.*** *A Lie group is a smooth manifold with compatible group structure. Its Lie algebra $\mathfrak{g} = T_eG$ encodes infinitesimal symmetry. But two Lie groups can share the same Lie algebra while differing globally: $SU(2)$ and $SO(3)$ are the fundamental example.*

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